

## ZS-F45

# The Seam Null-Power Route — A B3 Negative Audit: the Null-Power Reduction Theorem, with Three Reusable Tools

Kenny Kang — Z-Spin Cosmology Program (independent) · July 2026 · Foundations Series · Paper code **ZS-F45** · Version 1.4. B3 route-audit paper. Consumes ZS-M1, ZS-M3/F10, ZS-M44, ZS-M46, ZS-M47, ZS-F30–F43, ZS-A17, ZS-A23–A32.

**Verification: 47/47 PASS + 12/12 guards** (zs\_f45\_verify\_v1\_4.py; fail-closed, exits non-zero on any theorem-tier failure) | **5 firewalled observations** printed separately, never counted as PASS | **Zero fitted parameters** |  $(\mathbf{A}, \mathbf{Q}, \dim \mathbf{Z}) = (35/437, 11, 2)$  **LOCKED**.

### §0. Abstract

This is an *audit*, not a closure attempt. It opened the Planck-power / null-boundary route against barrier B3 as a candidate for the sole reopening gate F-F40.6, drove it to its honest terminus, and records the verdict: **the null-power attack reduces to exactly the same charge-unit residual  $\hat{e}_6 \equiv E\_len(seam)$  that ZS-F33/F42/F43 already localized** (Theorem F45.R). The route does not close B3. Its surviving value is (i) the audit verdict — a fourth, independent line reaches the same wall — and (ii) three reusable tools that outlive the negative verdict.

**Theorem F45.R (Null-Power Reduction).** Under the registered null-power postulate AX-PP and the (2+1) transplant, the value-carrying content of the null-power route reduces to the pre-existing  $\hat{e}_6 \equiv E\_len(seam)$  residual: the two value-gates G-PP2b and G-PP4 both terminate there — the first at the F-F42.36 charge-unit wall, the second at the ZS-M47 Parent-Factor residual — so the route supplies no new metric-side datum. [DERIVED; §3.]

**The three exported tools. (T1) Planck-Power/Frozen-Gas Identity** [IDENTITY]:  $\rho\_A = \eta\_Z \cdot L\_P / (c \cdot A\_d)$ ,  $A\_d = 4\pi(c/H\_d)^2$ , is an identity-level rewrite of the ZS-F42 IR ledger with  $\eta\_Z = (3/2)\Omega\_A, 0 = 249/242 = (1/4)c\_X\omega^2$  exactly ( $\omega^2$  cancels rationally), plus the two-ledger reading ( $N = 3 \Leftrightarrow \dim X$ : the IR ledger is the  $\hbar$ -free (G,c) ledger, unit  $L\_P$ , mirror to the G-free charge unit  $q\_P$ ). **(T2) Seam-as-Null-Energy Channel** [DERIVED-CONDITIONAL on KH1–KH4]: composing ZS-M46 Theorem C (GLW U(1)-current net) with the Morinelli–Tanimoto–Wegener null-plane HSMI theorem and the Ceyhan–Faulkner ANEC/QNEC theorem shows the seam generator  $\mathbf{P}$  is the averaged-null-energy operator of a null-cut inclusion — reusable by any future seam-energy argument. **(T3) A-vs- $\omega$  Holonomy Discriminator** [DERIVED]: in the ISO(2,1) Chern–Simons reading a deficit is a per-cycle *additive* increment, so of the corpus's two holonomy-like numbers only the telomere drift  $\mathbf{A}$  qualifies ( $d/dn[n\mathbf{A}] = \mathbf{A}$ ), while  $\omega = 2.2592$  is a constant reassigned to its ZS-M46 role as the helical twist  $\text{Re } \tau\_K = 0.35957$ .

**Barrier B3 is not closed; F-F40.6 armed, not fired;  $\Omega\_A, 0 = 83/121$  and  $U\_N$  untouched.** The absolute-scale question is handed to the sequestering execution ZS-F46 (radiative-stability face) and

stays, for its residual-value face, the documented B3-B frontier. Verification: 47/47 PASS + 12/12 guards; zero fitted parameters; (**A**, **Q**, dim **Z**) = (35/437, 11, 2) LOCKED.

### Epistemic Status Legend

| Tag                            | Meaning                                                                                                                                                                                              |
|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>IDENTITY</b>                | True by construction/definition; recorded so it is not mistaken for evidence.                                                                                                                        |
| <b>DERIVED</b>                 | Follows from corpus axioms/locked constants by explicit steps.                                                                                                                                       |
| <b>DERIVED-CONDITIONAL</b>     | Derived contingent on explicitly listed, falsifiable conditions.                                                                                                                                     |
| <b>IMPORTED-PROVEN</b>         | External theorem with published proof, consumed as-is (Deser–Jackiw–’t Hooft; Ashtekar–Varadarajan; Achúcarro–Townsend; Witten; Borchers; Ceyhan–Faulkner; Morinelli–Tanimoto–Wegener; Eckmann–Lim). |
| <b>NON-CLAIM</b>               | Explicit registration that a statement is not being made (here: the axiom AX-PP; the absolute value).                                                                                                |
| <b>OPEN / OPEN-TERMINAL</b>    | Well-posed and unresolved / unresolvable with current corpus tools.                                                                                                                                  |
| <b>NO-GO / CLOSED-NEGATIVE</b> | Proven impossibility / a route proven not to work under its pre-registered rule.                                                                                                                     |

### §1. The inherited terminus and the audit mandate

The B3 mechanism arc ended before this paper. ZS-F40 v1.2 closed all three determinant/clock routes negatively (the  $\sqrt{2}$  Refusal, the OPS exclusion, the  $\theta_1$  exclusion), recorded the mechanism programme **TERMINAL**, and left exactly one reopening trigger, **F-F40.6**, which demands a *new axiom-level input*. ZS-F41 supplied the lattice half as a candidate and installed the Terminal Stopping Rule. ZS-F42 delivered the frozen-gas hierarchy theorem and terminated at the **NON-CLAIM** that the membrane charge  $e_6$  cannot be fixed by (**A**, **Q**) and topology (gate F-F42.36). ZS-F43 re-typed the residual: by Stone–von Neumann–Mackey the charge unit is invisible from inside the Z-sector algebra, the well-posed target is the dimensionless  $\hat{e}_6$ , and **Theorem T1b** proved any successful execution lands at  $\hat{e}_6$  requiring exactly one metric-side datum.

The present paper asks one question: *does a null-boundary power principle on the seam supply that datum?* The answer, established through this paper's four theorems and executed gates, is **no** — **it reduces to the same  $\hat{e}_6$** . That negative answer, precisely located, is the paper's contribution, together with three tools produced along the way. The discipline is the predecessors': no value of  $\rho_\Lambda$ ,  $\hat{e}_6$ , or  $\ell$  is

evaluated; no depth mechanism is promoted; every load-bearing external ingredient is an imported theorem or explicitly tagged partial.

Notation:  $\hat{e}_6 := e_6/\bar{M}_P^2$ ,  $A_\partial = 4\pi(c/H_\partial)^2$  (Hubble-sphere convention). Terminology follows the corpus: *Z-sector* names the stage ( $\dim \mathbf{Z} = 2$ ), *Z-Spin* the action on it. All numerical claims are reproduced by `zs_f45_verify_v1_4.py` (47/47 PASS + 12/12 guards); scale-bearing numbers appear only in firewalled context. The full v1.0–v1.2 registration abstract is retained for provenance in Appendix D.

## §2. The three exported tools

**2.1 Tool T1 — the Planck-Power/Frozen-Gas Identity.** [IDENTITY / DERIVED.] Fix  $A_\partial = 4\pi(c/H_\partial)^2$  and  $L_P = c^5/G$ . Then  $\rho_\Lambda = \eta_Z \cdot L_P / (c \cdot A_\partial)$  reproduces the frozen-gas density  $\rho_\Lambda = 3\Omega_{\Lambda,0} \cdot \bar{M}_P^2 \cdot H_\partial^2$  iff  $\eta_Z = (3/2)\Omega_{\Lambda,0} = 3 \cdot 83 / (2 \cdot 11^2) = 249/242 = (1/4)c_\chi \omega^2$  (exact). The last equality is rational-exact:  $c_\chi \equiv 498/(121\omega^2)$  by the ZS-F42 pre-registration, so  $c_\chi \omega^2 = 498/121$  with  $\omega^2$  cancelling before any evaluation (checks A2–A3; numeric echo D7). The dimensional algebra  $L_P / (c \cdot A_\partial) = 2\bar{M}_P^2 H_\partial^2$  is verified with arbitrary units (check A4). The identity carries **no new empirical content** (the ZS-F43 §6.4 non-novelty pattern) — recorded as IDENTITY so it cannot be mistaken for evidence.

*Two-ledger corollary.* [Dimensional identity DERIVED; Z-Spin reading OBSERVATION.] The Planck force reads  $F_{pl} = G^\wedge\{2/(1-N)\} c^\wedge\{(5+N)/(N-1)\} \hbar^\wedge\{(3-N)/(1-N)\}$ ; the  $\hbar$ -exponent vanishes iff  $N = 3$  (checked  $N = 2 \dots 9$ , check A5), giving  $F_{pl} = c^4/G$ , unit  $c^5/G$  (check A6). So the IR frozen-gas ledger ( $\rho_\Lambda \propto H^2/G$ ,  $\hbar$ -free) is the *classical* ( $G$ ,  $c$ ) *ledger*, unit  $L_P$ , mirror to the *G-free quantum* charge unit  $q_P$  (ZS-F42 Cor. F42.11);  $\hbar$  crosses between them exactly once (ZS-F43 T4). That the  $\hbar$ -free ledger exists iff the spatial dimension equals  $\dim \mathbf{X} = 3$  is an alignment observation, not a derivation.

**2.2 Tool T2 — the Seam-as-Null-Energy Channel.** [DERIVED-CONDITIONAL on KH1–KH4.] The seam standard-pair generator  $\mathbf{P}$  is physically a null-energy operator, not an inert translation, by a three-theorem composition. (i) ZS-M46 Theorem C fixes the seam realization to the Guido–Longo–Wiesbrock  $U(1)$ -current net. (ii) Morinelli–Tanimoto–Wegener prove that for the free scalar the null-plane one-particle structure decomposes into lightlike fibres, the modular operator decomposes accordingly, and null-plane inclusions *are* half-sided modular inclusions whose translation generator is the integrated null energy. (iii) Ceyhan–Faulkner prove the half-sided generator is the averaged null energy and reproduces the shape derivative of relative entropy (QNEC-from-ANEC), given finite averaged null energy. Composing:  $\mathbf{P}$  is the averaged-null-energy operator of a null-cut HSMI (checks E1–E2). The residual is the model-fix release (KH1–KH4): the free-field realization is established; a corpus-intrinsic realization is the honest residual (check E3).

**2.3 Tool T3 — the A-vs- $\omega$  Holonomy Discriminator.** [DERIVED.] In the  $ISO(2,1)$  Chern–Simons formulation a point source's physical attribute is the rotation angle  $\sqrt{(w^2)}$  of its holonomy conjugacy class, which accumulates *additively* under loop composition (the Deser–Jackiw–'t Hooft rule adds conical rotations). A deficit is therefore a per-cycle *additive* increment. Of the corpus's two holonomy-like numbers, the derivative test is decisive (checks F1–F2):  $d/dn[n\mathbf{A}] = \mathbf{A}$  (additive  $\rightarrow$  deficit candidate),

while  $d/dk[2\pi k + \omega] = 2\pi$  with  $\omega$  dropping out (a constant, not an increment).  $\omega$  is already claimed elsewhere — it is the ZS-M46 helical internal twist  $\text{Re } \tau_K = \omega/2\pi = 0.35957$  (check F3). Hence  $A$  is the *unique* per-cycle rotation-part holonomy, and the identification  $A = 8\pi G_3 E_{\text{cycle}}$  is a derived selection, introducing no new constant (checks F4–F5).

### §3. The Null-Power Reduction Theorem (F45.R)

**3.1 Statement.** [DERIVED.] Under the registered null-power postulate AX-PP (per-tick seam power  $\leq c^5/4G$ , one tick =  $\hbar$ ; §4) and the (2+1) transplant (§3.4), the value-carrying content of the null-power route reduces to the pre-existing  $\hat{e}_6 \equiv E_{\text{len(seam)}}$  residual. Concretely, the two value-carrying gates both terminate at that single datum: **G-PP2b** at the F-F42.36 charge-unit wall, **G-PP4** at the ZS-M47 Parent-Factor residual. The route therefore supplies no new metric-side datum; it re-expresses the standing one in null-power language.

**3.2 G-PP2b terminates at the charge-unit wall.** [OPEN.] The (H-DEF) normalization  $A = 8\pi G_3 E_{\text{cycle}}$  splits into (N1) the geometric leg — Kaluza–Klein reduction  $G_3 = G_4/\ell_{\perp}$ , IMPORTED-PROVEN, giving  $E_{\text{cycle}} = A\ell_{\perp}/(8\pi G_4)$  (check G1a) — and (N2) the charge-quantum leg. (N2) is definitional: reconstructing  $8\pi G_3 E_{\text{cycle}}$  returns  $A$  identically (check G1b), with the per-cycle angle  $2\pi/N_{2\pi} = A$  by construction (check G1c). So (N2) does not derive  $E_{\text{cycle}}$ ; it defines it through  $A$ . An independent fix of  $E_{\text{cycle}}$  would fix  $\ell_{\perp}$ , hence the modulus  $\ell = M_K^{-1}$ , hence  $\hat{e}_6$  — exactly the charge unit F-F42.36 (and the ZS-F33 Charge-Unit Obstruction) proves cannot be fixed by  $(A, Q)$  and topology. **G-PP2b is OPEN: the geometric leg is proven, the normalization leg is the charge-unit wall in new coordinates** (check G1d). No  $\ell_{\perp}$ ,  $E_{\text{cycle}}$ , or  $\hat{e}_6$  value is evaluated (guard G12).

**3.3 G-PP4 terminates at the Parent-Factor residual.** [OPEN-TERMINAL.] A Wieland-type discrete→continuous transition of the tick-power  $e^{\{2\pi t\}}$  rescale at  $t^* = Q$  is not constructible with current tools: the corpus discrete/continuum distinction is controlled by *sector dimension* (ZS-M6 §5.5 — X-sector truncated octahedron tiles  $\mathbb{R}^3$ , continuum emerges; Y-sector truncated icosahedron and  $\dim Z = 2$  do not tile, spectra discrete), **not by depth**;  $t^* = Q$  is a depth *selection* (ZS-F38), not a transition *generator* (checks H1–H2). The residual is the ZS-M47 Parent-Factor Realization Problem, identified by ZS-F43 §4.3 with the  $E_{\text{len(seam)}}$  OPEN. Partial positive: the mediation channel rank  $\leq \dim Z = 2$  (ZS-Q7/T12) gives an  $\ln 2$  information ceiling on any transition threshold (check H3). **G-PP4 is OPEN-TERMINAL — reopenable only through the ZS-M47 Parent-Factor Realization, i.e. the same axiom-level input F-F40.6 already names.**

**3.4 The (2+1) transplant re-bases the value leg on a theorem — but its normalization is exactly G-PP2b.** [general IMPORTED-PROVEN; instance DERIVED-CONDITIONAL on (H-DEF)  $\wedge$  (H-TICK).]  $\dim Z = 2$  makes the seam world-volume (2+1)-dimensional, where the contested 3+1 maximum-luminosity conjecture is *not needed*: by Deser–Jackiw–’t Hooft and Ashtekar–Varadarajan a point source carries deficit  $\alpha = 8\pi G_3 m$  with  $\Sigma\alpha \leq 2\pi$ , hence  $M \leq 1/(4G_3)$ ; by Achúcarro–Townsend and Witten the (2+1) theory is Chern–Simons, so deficit *is* rotation-holonomy — the phase-vs-metric gap closes by theorem in the seam's own dimension. Under (H-DEF) (tool T3), the cumulative deficit reaches  $2\pi$  at  $N_{2\pi} = 2\pi/A = 874\pi/35 = 78.4500565\dots$  cycles — the corpus's frozen winding rule — with  $\Sigma E =$

$1/(4G_3)$ : **the winding change is bound saturation** (check C2). Under (H-TICK) ( $G_3 = G_4/\ell_{\perp}$ ,  $t_{\text{tick}} = \ell_{\perp}/c$ ) the per-tick bound is  $\mathbf{P_{max}} = c^5/(4G_4)$  with  $\ell_{\perp}$  cancelling identically (checks C3a–C3b), the coefficient  $1/4$  imported. The rate leg costs nothing: the Borchers relation  $\Delta^{\{it\}}U(a)\Delta^{\{-it\}} = U(e^{\{\mp 2\pi t\}}a)$  is exact in ZS-M46 (check B4) and HSMI saturate the modular-chaos bound. **But the transplant's normalization is exactly G-PP2b (§3.2)** — the theorem re-bases the *epistemic status* of the value leg (conjecture  $\rightarrow$  theorem) without moving its terminus, which remains the charge-unit wall. This is why F45.R holds: the strongest available formulation of the null-power value leg still reduces to  $\hat{e}_6$ .

**3.5 The audit verdict.** [DERIVED.] Both value-gates reduce to the same single dimensionful residual  $\hat{e}_6 \equiv E_{\text{len}}(\text{seam})$  that ZS-F43 localized. The null-power route is thus a *fourth* independent line — after the determinant/clock terminus (F40), the frozen-gas terminus (F42), and the unit-invisibility terminus (F43) — reaching the same wall. Per the deep-exploration protocol, a convergent search whose value-carrying nodes remain OPEN confirms "the residual cannot be closed with current corpus tools," and a fourth route landing on the same terminus is positive structural evidence that the terminus is real, not a single-tool artifact.

#### §4. The registered axiom AX-PP (retained as NON-CLAIM)

[NON-CLAIM — registration only.] For provenance and for downstream gates, the candidate datum audited above is the two-clause postulate: **(PP-1)** the per-tick power transported by Z-Spin mediation through the seam null boundary is bounded by  $c^5/(4G)$ , with the register's topological phase transition at saturation; **(PP-2)** one register tick carries one quantum of action  $\hbar$ . Consequence (bookkept, not evaluated):  $\text{PP-1} \wedge \text{PP-2}$  give  $t_{\text{tick}} = 2t_P$  (check C5), the  $O(1) = 2$  from the imported  $1/4$ ;  $\hbar$  enters the sector exactly once, realizing the ZS-A25 "exactly one dimensionful datum" no-go. AX-PP is **not** a claim that B3 is closed, **not** a derivation of  $e_6$  (F-F42.36 respected), **not** a promotion of the depth mechanism (guard G5). The audit (§3) shows AX-PP, even granted, supplies no new datum — which is precisely why the route is recorded negative.

#### §5. Falsification gates and ledger

Table 1. Gate statuses after the audit.

| Gate                                                         | Status                                         |
|--------------------------------------------------------------|------------------------------------------------|
| <b>G-PP1</b> (seam $\mathbf{P}$ = null-energy operator)      | DERIVED-CONDITIONAL on KH1–KH4 (tool T2)       |
| <b>G-PP2a</b> ((H-DEF) discriminator: $\mathbf{A}$ selected) | DERIVED (tool T3)                              |
| <b>G-PP2b</b> ((H-DEF) normalization)                        | <b>OPEN</b> — F-F42.36 charge-unit wall (§3.2) |
| <b>G-PP3</b> (clock discharge via ZS-F39 frozen rule)        | registered; not evaluated (risk R5)            |

| Gate                                        | Status                                        |
|---------------------------------------------|-----------------------------------------------|
| <b>G-PP4</b> (depth transition at $t^*=Q$ ) | <b>OPEN-TERMINAL</b> — ZS-M47 residual (§3.3) |
| <b>G-PP5</b> (C_UV band; automatic)         | evaluated only after G-PP2a/b                 |
| <b>G-PP6</b> (scope honesty; U_N untouched) | active                                        |

Table 2. Reduction ledger.

| Item                                                                                             | Status                                                             |
|--------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|
| $\eta_Z = 249/242 = (1/4)c_\chi\omega^2$ identity (T1)                                           | <b>IDENTITY / DERIVED</b>                                          |
| seam <b>P</b> = averaged-null-energy operator (T2)                                               | <b>DERIVED-CONDITIONAL on KH1–KH4</b>                              |
| (H-DEF) discriminator — <b>A</b> unique (T3)                                                     | <b>DERIVED</b>                                                     |
| (2+1) chain (DJt'H $\times$ CS $\times$ Gauss–Bonnet)                                            | <b>IMPORTED-PROVEN</b>                                             |
| saturation identity; $P_{\max} = c^5/4G_4$                                                       | <b>DERIVED-CONDITIONAL on (H-DEF) <math>\wedge</math> (H-TICK)</b> |
| <b>Theorem F45.R: route reduces to <math>\hat{e}_6 \equiv E_{\text{len}}(\text{seam})</math></b> | <b>DERIVED (§3)</b>                                                |
| AX-PP (PP-1, PP-2)                                                                               | <b>NON-CLAIM</b>                                                   |
| barrier B3                                                                                       | <b>NOT closed</b> ; F-F40.6 armed, not fired                       |
| $\Omega_{\Lambda,0} = 83/121$ ; U_N                                                              | <b>untouched</b>                                                   |

## §6. Conclusion

The null-power route is a clean negative result: audited to its terminus, it reduces to the same charge-unit residual  $\hat{e}_6 \equiv E_{\text{len}}(\text{seam})$  that three prior routes already reached (Theorem F45.R). What survives is not a datum but three reusable tools — the  $\eta_Z = 249/242$  identity, the seam-as-null-energy channel, and the A-vs- $\omega$  discriminator — plus the structural fact that under (H-DEF) the corpus's frozen winding rule  $N_{2\pi} = 2\pi/A$  is the Deser–Jackiw–'t Hooft saturation condition. The absolute-scale question passes to the sequestering execution ZS-F46, which addresses the radiative-stability face while leaving the same residual as a calibrated boundary datum. B3 is not closed; F-F40.6 armed, not fired;  $\Omega_{\Lambda,0} = 83/121$  and U\_N untouched.

## Acknowledgements & Code Availability

Verification code: zs\_f45\_verify\_v1\_4.py (Python 3; mpmath 50 dps; exact Fraction arithmetic; SymPy; NumPy). Fail-closed; prints the firewalled block under an explicit banner, never counting it as PASS. Result: 47/47 PASS + 12/12 guards; 5 firewalled observations. Developed under the corpus session protocol with multi-AI adversarial review for circularity detection.

#### Appendix A. Numerical dictionary (reproduced digits)

| Quantity                                           | Value                                | Check     |
|----------------------------------------------------|--------------------------------------|-----------|
| $\mathbf{A}; \mathbf{Q}; (\dim X, Z, Y)$           | 35/437; 11; (3,2,6)                  | LOCKED    |
| $\omega; \kappa_\lambda;  \lambda^* $              | 2.2592495540; 0.1148346250; 0.8915   | K1–K3     |
| $\Omega_\Lambda, 0; 3\Omega_\Lambda, 0$            | 83/121; 249/121                      | D6        |
| $c_\chi = 498/(121\omega^2)$                       | 0.8063350941                         | A1        |
| $\eta_Z = 249/242$                                 | <b>1.0289256198</b>                  | A2–A4, D7 |
| $\ \Pi - D\ _F/\sqrt{Q}$ (any diagonal D)          | $\sqrt{2}$                           | B1–B2     |
| $N_{2\pi} = 2\pi/A = 874\pi/35$                    | 78.4500565496                        | C1        |
| $\Sigma E(N_{2\pi})$                               | $1/(4G_3)$                           | C2        |
| $P_{\max}; \partial P_{\max}/\partial \ell_\perp$  | $c^5/(4G_4); 0$                      | C3a–C3b   |
| $t_{\text{tick}} (PP\text{-}1 \wedge PP\text{-}2)$ | $2t_P$                               | C5        |
| $d/dn[n\mathbf{A}]; d/dk[2\pi k + \omega]$         | $\mathbf{A}; 2\pi$ ( $\omega$ drops) | F1–F2     |
| $\omega/2\pi = \text{Re } \tau_K$                  | 0.3595707342                         | F3        |
| $\hat{e}_6 = 2\pi e^{\{-4\pi\mathbf{Q}\}}$         | $5.829 \times 10^{-60}$              | D4        |

Firewalled (never PASS):  $H_{\partial/\tilde{M}_P} = 5.9009 \times 10^{-61}$  (O-1);  $C_{UV} = 1.244$  (O-2);  $GW150914/\text{bound} = 4.0 \times 10^{-3}$  (O-4);  $\sqrt{e_6} = 5.879 \text{ meV}$  (O-5).

#### Appendix B. Deep-exploration record (condensed)

The value-gate execution (Records 3–5 of the v1.0–v1.2 line) converged  $5 \rightarrow 3 \rightarrow 1 \rightarrow 0$  to genuine OPEN: G-PP2b to the F-F42.36 charge-unit wall, G-PP4 to the ZS-M47 residual, both reducing to  $\hat{e}_6$ . The self-reference check refused the over-read "seam = null energy  $\Rightarrow$  closed": G-PP1 is held to DERIVED-CONDITIONAL because the ZS-M46 realization is model-fixed; the  $\gamma = \sqrt{3}$  numerological

temptation is blacklisted (guard G4). The elevation route (external PROVEN mathematics: Morinelli–Tanimoto–Wegener; Ceyhan–Faulkner; Deser–Jackiw–’t Hooft; ISO(2,1) Chern–Simons) supplies the tools T2–T3, not a new axiom. Full node-by-node records are preserved in the v1.2 archive.

### Appendix C. Dependency table

No upstream value or status is changed by this paper (guard G6; re-audited against The Book v11.0). Consumed: ZS-M1 ( $z^*$ , PROVEN); ZS-M46 v1.5 (standard pair,  $P \geq 0$ , Borchers, Thm C; PROVEN/DERIVED-CONDITIONAL); ZS-M47 v2.0 (Parent-Factor Realization, OPEN); ZS-F38 v1.2 (Q-chain,  $t^* = Q$ ); ZS-F40 v1.2 (terminal verdict;  $\sqrt{2}/OPS/\theta_1$ ; F-F40.6); ZS-F42 v1.9 ( $c_\chi$ ; F-F42.36); ZS-F43 v1.1 (T1b,  $\hat{e}_6$ , one metric-side datum); ZS-A25 v1.6 ( $[\Lambda, \hat{T}_4] = i\hbar$ ); ZS-A30 v2.1 ( $\Omega_\Lambda, 0 = 83/121$ ); ZS-A32 v1.1 (MC  $p = 0.50\%$ , firewalled).

### Appendix D. Provenance — the v1.0–v1.2 registration abstract (retained verbatim)

*The following is the original registration abstract, retained unedited for provenance; its claims are superseded in framing (registration  $\rightarrow$  audit) but not in content by §0.*

ZS-F40 v1.2 recorded the B3 determinant/clock mechanism programme TERMINAL and left exactly one reopening gate, F-F40.6, which demands a new axiom-level input. ZS-F43 v1.1 then proved (Theorem T1b) that any successful execution necessarily lands at the dimensionless  $\hat{e}_6$  and requires exactly one metric-side datum. This paper does not fire the gate. It registers a candidate for that datum — a null-boundary power principle on the Z-Spin seam — in four theorems: T1 (the Planck-power/frozen-gas identity  $\eta_Z = 249/242 = (1/4)c_\chi\omega^2$ , IDENTITY/DERIVED, with the  $N = 3 \Leftrightarrow \dim X$  two-ledger classification); T2 (the CRT-4a insertion point and  $\sqrt{2}/OPS/\theta_1$  disjointness, DERIVED); T3 (the Deser–Jackiw–’t Hooft (2+1) transplant — general IMPORTED-PROVEN, instance DERIVED-CONDITIONAL on (H-DEF)  $\wedge$  (H-TICK) — with the saturation identity  $N_2\pi = 2\pi/A \leftrightarrow \Sigma E = 1/4G_3$  and the  $\ell_\perp$ -cancelling per-tick bound  $P_{\max} = c^5/4G$ ); and T4 (the registered two-clause axiom, NON-CLAIM, with consequence  $t_{\text{tick}} = 2t_P$ ). v1.1 discharged G-PP1 (seam  $P =$  averaged-null-energy via M46.C  $\circ$  Morinelli–Tanimoto–Wegener  $\circ$  Ceyhan–Faulkner; DERIVED-CONDITIONAL) and G-PP2a ((H-DEF) A-vs- $\omega$  discriminator;  $A$  selected uniquely; DERIVED). v1.2 executed the value gates: G-PP2b OPEN (definitional normalization  $\rightarrow$  F-F42.36 wall) and G-PP4 OPEN-TERMINAL (sector-dimensional, not depth-indexed; = ZS-M47 residual), both reducing to the same  $\hat{e}_6$ . Barrier B3 NOT closed; F-F40.6 armed, not fired;  $\Omega_\Lambda, 0 = 83/121$  and  $U_N$  untouched.

### Version History

**v1.4 (July 2026):** Compression + consolidation of the audit. No theorem, gate, or verification result changed; the paper is reduced  $\sim 40\%$  from v1.3. Changes of record: (i) the audit framing is now the whole paper, with the central result named **Theorem F45.R (Null-Power Reduction)** — the route reduces to the pre-existing  $\hat{e}_6 \equiv E_{\text{len}}(\text{seam})$  residual; (ii) the full v1.0–v1.2 registration abstract is moved from body §0.1 to **Appendix D** (provenance), restoring reading flow; (iii) body §1–§6 condensed, the three exported



tools foregrounded in §2, the value-gate executions summarized in §3; (iv) the verification file is copied to **zs\_f45\_verify\_v1\_4.py** (header updated to v1.4; content identical to the v1.2 suite — 47/47 PASS + 12/12 guards), removing the v1.3 filename mismatch. Re-audited against The Book v11.0; no upstream value or status moved. Barrier B3 NOT closed; F-F40.6 armed, not fired;  $\Omega_{\Lambda,0} = 83/121$  and  $U_N$  untouched.

**v1.3 (July 2026):** Repositioning from "registration" to "audit"; three reusable tools foregrounded; hand-off to ZS-F46 directed. (Full v1.0–v1.2 history preserved in the v1.2 archive: v1.0 registration of AX-PP in four theorems; v1.1 discharge of G-PP1/G-PP2a; v1.2 execution of the value gates to genuine OPEN.)